

Microsoft Azure AI Foundry

Complete Architecture & Component Hierarchy

2025 / 2026 · All Components · GA & Preview Status

 Resource	 Projects	 Agents	 Tools	 Dev Exp	 Security
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① Foundry Resource — Top-Level Azure Resource

The Foundry Resource is the root Azure resource (Microsoft.CognitiveServices/accounts). Created in the Azure Portal. All projects, model deployments, and connections are scoped under it.

1.1 Model Deployments

- └─ **OpenAI Family**
 - └─ **GPT-5 / GPT-5-chat** *Latest flagship model* GA
 - └─ **GPT-4o / GPT-4.1 / GPT-4.1-nano** *1M token context on GPT-4.1* GA
 - └─ **o3 / o4-mini** *Reasoning models* GA
 - └─ **DALL·E / GPT-image-1** *Image generation* GA
 - └─ **Whisper / gpt-4o-transcribe** *Audio / speech-to-text* GA
- └─ **Microsoft Family**
 - └─ **Phi-4** *Small language model* GA
 - └─ **Phi-3.5** *Lightweight SLM* GA
- └─ **Partner & Community**
 - └─ **Meta Llama 3.1 / 3.2** *Open-source*
 - └─ **Mistral Large / NeMo** *Document AI strength*
 - └─ **Cohere Command R+ / Rerank v4** *Enterprise search*
 - └─ **xAI Grok 3 / Grok 4 Fast**
 - └─ **DeepSeek / Kimi / Moonshot**
 - └─ **10,000+ Hugging Face open-source models**
- └─ **Model Router** *Auto-routes to best model per call* GA
Balances cost / latency / capability · Version: 2025-11-18

1.2 Deployment Types

- └─ **Global Standard** *Shared quota, flexible — default choice*
- └─ **Provisioned Throughput (PTU)** *Reserved capacity — predictable performance*

- └─ **Serverless Endpoint (MaaS)** *Pay-per-token — no reserved capacity*
- └─ **Managed Compute** *Custom hosting — Hub-based projects only*

1.3 Security Settings

- └─ **Managed Identity** *System-assigned or user-assigned*
- └─ **RBAC Roles**
 - └─ **Foundry Account Owner** *Full control of the resource*
 - └─ **Foundry Owner** *Project-level full control*
 - └─ **Foundry User** *Project-level read/use*
 - └─ **Foundry Project Manager** *Manage project members*
- └─ **Network / Private Endpoints** *VNet injection, firewall rules*

1.4 Connections (Pointers to External Services)

Connections authenticate and consume Microsoft and third-party services within Foundry projects. Required for Standard Agents and knowledge tools.

Azure Services

- └─ → **Azure AI Search** GA · *Required for Standard Agent*
Vector + semantic search indexes · RAG grounding
- └─ → **Azure Storage** GA · *Required for Standard Agent*
Blobs, files, documents
- └─ → **Azure Cosmos DB** Preview · *Code-only (no portal UI)*
Required for Standard Agent · Agent state + threads
- └─ → **Azure OpenAI** GA · *Your own Azure OpenAI resource*
GPT-5, GPT-4o, Embeddings with Azure security
- └─ → **Application Insights** GA
Performance monitoring · Trace + diagnose agent runs
- └─ → **Azure Key Vault** GA · [!\[\]\(c6a8736a601a632e2c96605cf66055ed_img.jpg\)](#) *Deleting KV breaks Foundry resource*
Secrets, keys, certificates · BYO Vault supported
- └─ → **Azure Databricks** Preview · *Code-only*
Access Databricks Jobs + Genie Spaces at runtime
- └─ → **Azure APIM** Preview · *Code-only*
API governance for AI models called in Agent Service

Microsoft Services

- └─ → **SharePoint** Preview · *Code-only*
Document storage + collaboration · Agents read/manage org documents
- └─ → **Microsoft Fabric** Preview · *Code-only*
AI Skills — conversational Q&A on Fabric data
- └─ → **Foundry (another resource)** GA
Connect to OTHER Foundry resources · Cross-resource model sharing

Search & Web Grounding

- └─ → **Grounding with Bing Search** GA
Real-time web data · Current results in agent responses
- └─ → **Grounding with Bing Custom Search** Preview · *Code-only*
Tailored / scoped web search

- └─ → **Serp** GA
Search Engine Results Pages · Real-time SERP data

External & Custom

- └─ → **OpenAI (direct — not Azure)** GA
Connect to openai.com directly · Non-Azure OpenAI account
- └─ → **Serverless Model** Preview · Code-only
Serverless API deployment connections
- └─ → **Model Gateway** Preview · Code-only
Governance for AI model calls in Agent Service
- └─ → **API Key** GA
Any external service with an API key · Per-target auth
- └─ → **Custom Key** GA
Store keys + metadata for any target · LangChain scenarios · You manage auth

Connection creation methods:

☑ Created in Portal UI	🔑 Code-Only (Bicep / SDK)
AI Search · Storage · OpenAI · Key Vault App Insights · Bing · Serp · API Key Custom Key · Foundry · OpenAI (direct)	Cosmos DB · SharePoint · Fabric Databricks · APIM · Model Gateway Serverless Model · Bing Custom Search

② Foundry Project — Recommended (New Gen)

The primary development workspace. Supports the latest Agent Service, project endpoints, and the unified azure-ai-projects SDK v2.

2.1 Agent Service

Agent Types

- └─ **Prompt Agents** Instruction-based · Stateless
- └─ **Hosted Agents** Deployed workflows · Long-running · Stateful

Agent Core Components

- └─ **System Prompt / Instructions** Defines agent behavior and persona
- └─ **Model Deployment** Which model the agent uses
- └─ **Threads** Conversation sessions
- └─ **Runs** Execution instances per thread
- └─ **Steps** Individual actions within a run

Agent Tools

- └─ **Azure AI Search tool** GA · Vector + semantic + hybrid search
- └─ **Bing Web Search tool** GA · Real-time internet grounding
- └─ **Function Calling tool** GA · Custom business logic + APIs
- └─ **Code Interpreter tool** GA · Run Python in sandbox · Data analysis
- └─ **File Search tool** GA · Upload + chunk + in-session RAG

- └─ **MCP Server tool** *Preview · mcp.ai.azure.com · 1,400+ systems*
- └─ **Browser Automation** *Preview · Computer Use · Web task automation*
- └─ **A2A Protocol tool** *Preview · Agent-to-Agent calls*

Memory Store (Preview)

- └─ **Long-term memory across sessions**
- └─ **Automatic extraction + consolidation**
- └─ **Free during preview · Pay only for model calls**

Multi-Agent Workflows

- └─ **Visual builder in Foundry portal**
- └─ **Agent-to-Agent (A2A) calls**
- └─ **Connected Agents · Sub-agent delegation**

2.2 Playground

- └─ **Chat Playground** *Test LLM conversations · Zero code*
- └─ **Agents Playground** *Test your agents end-to-end*
- └─ **Image Playground** *DALL·E / GPT-image-1 · Image generation*
- └─ **Audio Playground** *Whisper / TTS · Speech testing*
- └─ **Compare Mode** *Side-by-side model output comparison*

2.3 Evaluations

- └─ **Groundedness score** *Is the answer based on real/grounded data?*
- └─ **Relevance score** *Is the answer on-topic?*
- └─ **Coherence score** *Does the answer make logical sense?*
- └─ **Fluency score** *Is the language natural and readable?*
- └─ **Safety score** *Harmful content detection*
- └─ **Task Completion score** *Multi-turn goal achievement*
- └─ **Custom Evaluators** *Define your own scoring metrics*

2.4 Observability (GA as of Build 2026)

- └─ **Trace**
 - └─ **End-to-end telemetry for every step**
 - └─ **Prompt → model call → tool → sub-agent hop**
 - └─ **Any framework: LangChain, AutoGen, OpenAI SDK**
 - └─ **OpenTelemetry (OTel) exporter · Point and go**
- └─ **Evaluate (in production)**
 - └─ **Single-turn quality scoring**
 - └─ **Multi-turn conversation scoring**
- └─ **Monitor**
 - └─ **Real-time issue detection**
 - └─ **Alerts + dashboards**
 - └─ **→ Azure Monitor (external)**

- └─ **Optimize**
 - └─ **Production signal analysis**
 - └─ **Ranked evidence-backed improvement suggestions**

2.5 Fine-tuning Jobs

- └─ **LoRA** *Lightweight adapter fine-tuning*
- └─ **QLoRA** *Quantized LoRA · Lower memory*
- └─ **DPO** *Direct Preference Optimization*
- └─ **Developer Tier** *No hosting fee to experiment*

2.6 Files & Data

- └─ **Uploaded Files** *In-session RAG · Chunked automatically*
- └─ **Vector Indexes** *Chunked + embedded document indexes*
- └─ → **Azure Blob Storage** *(external) · Actual files stored here*

2.7 Project Endpoint

- └─ **REST API (GA: 2025-05-01)** *Your .NET / Python / JS app calls this*
Set PROJECT_ENDPOINT + MODEL_DEPLOYMENT_NAME as environment variables

③ Hub-Based Project — Classic / AML-Based

Legacy project type built on Azure Machine Learning workspace. Still supported. Use for Prompt Flow pipelines, full fine-tuning, or AML-based workloads.

3.1 Prompt Flow

- └─ **Standard Flow** *General LLM pipeline · DAG-based*
- └─ **Chat Flow** *Conversational LLM pipeline*
- └─ **Evaluation Flow** *Quality testing pipeline*

3.2 Other Capabilities

- └─ **Fine-tuning** *Same options as Foundry Project*
- └─ **Evaluations** *Same options as Foundry Project*
- └─ **Managed Compute** *AML clusters · GPU/CPU*
- └─ **Connections** *Storage / KV / AppInsights · Same types*

3.3 Auto-Created Resources (on Hub creation)

- └─ **Azure Storage Account**
- └─ **Azure Key Vault**
- └─ **Azure Application Insights**

└─ **Azure Container Registry**

⚠ Deleting any of these breaks the Hub. AzureML SDK v1 EOL: June 30, 2026 — migrate to SDK v2.

④ Foundry Tools — Prebuilt AI (formerly Azure AI Services)

Call via API. No model selection needed. Ready-to-use AI capabilities covering Speech, Vision, Language, Documents, and Safety.

4.1 Speech

- └─ **Speech-to-Text (STT)** *Whisper + gpt-4o-transcribe / mini-transcribe*
- └─ **Text-to-Speech (TTS)** *Neural voices · High naturalness*
- └─ **Real-time Translation** *Streaming audio translation*

4.2 Vision

- └─ **Image Analysis** *Describe, tag, caption images*
- └─ **OCR** *Read text from images and documents*
- └─ **Face API** *Detection + verification*
- └─ **Object Detection** *Locate and identify objects in images*

4.3 Language

- └─ **Sentiment Analysis** *Positive / negative / neutral + confidence*
- └─ **Named Entity Recognition (NER)** *People, orgs, locations, dates*
- └─ **Key Phrase Extraction** *Extract main topics from text*
- └─ **PII Detection** *GA: 2026-05-01 · Password, PIN, Zip, Airport*
- └─ **Language Detection** *Identify language from text*
- └─ **Summarization** *GA: 2025-06-10 · Phi-based model · Issue & Resolution*
- └─ **Custom Question Answering (CQA)** *Deploy as agent from Foundry portal*
- └─ **Conversational Language Understanding (CLU)** *Intent + entity extraction*

4.4 Document AI

- └─ **Form Recognizer** *Extract structured fields from forms*
- └─ **Layout Analysis** *Tables, paragraphs, reading order*
- └─ **Prebuilt Models** *Invoices · Receipts · W9s · IDs*

4.5 Translator

- └─ **Real-time Translation** *100+ languages*
- └─ **Batch Document Translation** *Large-scale document translation*

4.6 Content Safety

- └─ **Harmful Content Filter** *Violence, hate, self-harm detection*
- └─ **Jailbreak Detection** *Prompt injection + adversarial attack detection*
- └─ **Groundedness Detection** *Is the response factually anchored?*

⑤ Developer Experience

How developers access, build, and deploy with Azure AI Foundry. Multiple surfaces from portal to CLI.

5.1 Foundry Portal

- └─ **ai.azure.com** *Visual workspace for all capabilities*
Browse models · Test in playground · Build agents · Run evaluations · Fine-tune

5.2 Foundry SDK (azure-ai-projects v2)

- └─ **Agents** *Build and run agents*
- └─ **Inference** *Call deployed models*
- └─ **Evaluations** *Score outputs programmatically*
- └─ **Memory** *Long-term memory management*
- └─ **Python** *pip install azure-ai-projects*
- └─ **.NET / C#** *NuGet: Azure.AI.Projects*
- └─ **JavaScript / TypeScript** *npm install @azure/ai-projects*
v2.0.0b3+ — unified package (azure-ai-agents dependency removed)

5.3 REST API

- └─ **GA version: 2025-05-01** *Direct HTTP calls to project endpoint*

5.4 VS Code Extension — Foundry Toolkit

- └─ **Browse + Deploy Models** *From Model Catalog directly*
- └─ **Build + Deploy Hosted Agents** *Foundry Toolkit: Deploy Hosted Agent*
- └─ **Open in VS Code from Portal** *One-click import of agent + endpoint + code*
- └─ **Generate Sample Code** *For any agent or model configuration*
Install: VS Code Marketplace → search 'Azure AI Foundry' or 'Microsoft Foundry Toolkit'

5.5 Azure Developer CLI (azd)

- └─ **azd ai agent init** *Scaffold project · Bicep IaC · azure.yaml*
- └─ **azd up** *Provision + deploy · Typically 5-10 min*
Managed Identity configured automatically · RBAC + endpoint security via IaC templates

5.6 Foundry Local

- └─ **Run models on-device** *No cloud required*

- └─ **Windows NPU / CPU** *Intel, AMD, Qualcomm NPU support*
- └─ **CLI + SDK** *Local inference for IoT / Edge / Mobile*

⑥ Security & Governance — Spans Everything

Security and governance apply across the entire Foundry Resource, all Projects, and all connected services.

- └─ **Microsoft Entra ID**
 - └─ **Identity + login** (OAuth2 / OIDC)
 - └─ **RBAC role assignments**
 - └─ **Conditional Access policies**
- └─ **Azure Key Vault**
 - └─ **Secrets, keys, certificates**
 - └─ **BYO Vault support**
- └─ **Private Endpoints + VNet**
 - └─ **Network isolation for enterprise**
 - └─ **Azure Firewall / DDoS protection**
 - └─ **Hub-spoke topology support**
- └─ **Azure Policy**
 - └─ **Governance rules + compliance**
 - └─ **Resource tagging + locks**
- └─ **Azure Monitor + Diagnostics**
 - └─ **Platform-level logs**
 - └─ **Cost Management + unified billing dashboard**
- └─ **Responsible AI**
 - └─ **Content Safety filters** (built into all model calls)
 - └─ **Fairness evaluation**
 - └─ **Transparency reports**

⑦ GA vs Preview Status — 2025/2026

Component	Status	Notes
Agent Service	☑ GA	Generally available
Tracing & Evaluations	☑ GA	Reached GA at Build 2026
Model Router	☑ GA	Version 2025-11-18
PII Detection (Language)	☑ GA	API version 2026-05-01
Summarization (Language)	☑ GA	Phi-based, GA: 2025-06-10

REST API	☑ GA	Version 2025-05-01
Foundry SDK v2	🟡 Beta	v2.0.0b3 · Stabilizing
Memory Store	🟡 Preview	Free during preview
Browser Automation / Computer Use	🟡 Preview	Agentic web automation
A2A Protocol Tool	🟡 Preview	Agent-to-Agent calls
Foundry MCP Server	🟡 Preview	mcp.ai.azure.com (Dec 2025)
Cosmos DB Connection	🟡 Preview	Code-only creation
SharePoint Connection	🟡 Preview	Code-only creation
Fabric Connection	🟡 Preview	Code-only creation
Databricks Connection	🟡 Preview	Code-only creation
APIM Connection	🟡 Preview	Code-only creation

⑧ Key Rules Summary

Category	Details
Foundry OWNS	Models · Agents · Playground · Evaluations · Fine-tuning · Foundry Tools · Observability
Foundry CONNECTS TO	AI Search · Storage · Cosmos DB · Key Vault · App Insights · SharePoint · Fabric · Bing · Databricks · APIM + 11 more connection types
Foundry DOES NOT HOST	Your .NET app (→ App Service) · Your database (→ Cosmos DB / SQL) · Your SharePoint · Your pipeline logic (→ ADF)
Portal vs Portal	portal.azure.com = Create & manage Azure resources (IT/DevOps) ai.azure.com = Build AI apps and agents (Developers)
Foundry Project vs Hub	Foundry Project = new gen, recommended, Agents GA Hub-Based = classic AML, Prompt Flow, AzureML SDK v1 (EOL June 2026)
Connections rule	11 connection types creatable in Portal UI 8 connection types require code-only (Bicep / SDK)
Billing	Foundry, AI Search, Storage, Cosmos DB all bill SEPARATELY in your Azure subscription

⑨ Quick Reference Links

Resource	URL
Foundry Portal	https://ai.azure.com
Official Docs	https://learn.microsoft.com/azure/ai-foundry

Architecture Concepts	https://learn.microsoft.com/azure/ai-foundry/concepts/architecture
Connections How-To	https://learn.microsoft.com/azure/foundry/how-to/connections-add
Agent Service Docs	https://learn.microsoft.com/azure/ai-foundry/agents
Baseline Chat Arch	https://learn.microsoft.com/azure/architecture/ai-ml/architecture/baseline-microsoft-foundry-chat
SDK Install (Python)	<code>pip install azure-ai-projects</code>
SDK Install (.NET)	<code>dotnet add package Azure.AI.Projects</code>
azd CLI Quickstart	https://learn.microsoft.com/azure/developer/azure-developer-cli